

DATA STRUCTURE

LAB PROGRAMS

8. Write a C program to search a number using Linear Search method

```
#include <stdio.h>
```

```
int main() {
```

```
    int arr[100], n, key, i, found = 0;
```

```
    printf("Enter number of elements: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter array elements:\n");
```

```
    for(i = 0; i < n; i++) {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    printf("Enter number to search: ");
```

```
    scanf("%d", &key);
```

```
    for(i = 0; i < n; i++) {
```

```
        if(arr[i] == key) {
```

```
            found = 1;
```

```
            printf("Element found at position %d", i + 1);
```

```
            break;
```

```
    }  
}  
  
if(found == 0) {  
    printf("Element not found");  
}  
  
return 0;  
}
```

```
Enter number of elements: 5  
Enter array elements:  
3 5 6 4 5  
Enter number to search: 5  
Element found at position 2  
-----  
Process exited after 12.06 seconds with return value 0  
Press any key to continue . . .
```